

Rigour Without Infrastructure

Three Propositions on Claim-Card Discipline as a Substitute for Institutional Certification

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In brief. Problem. A person with no university, no lab, and no research team who now has AI assistance faces a new failure mode: text that reads as smart and academic while it stays unclear which part is evidence, which part is interpretation, and which part AI smoothed over past what the data supports (??). **Proposition.** Three candidate hypotheses (??) — a design mechanism (H1), a conceptual claim about where rigour lives (H2), and an empirical claim about cross-vendor AI checking (H3) — propose that a claim-card discipline bound to Existence–Attribution–Disclosure and gated by an independence ladder can make a standalone scholar’s claim reader-checkable without an institutional credential. **Method.** Each proposition carries a named falsifier and is placed against 48 independently verified citation cards in a conversation-with-the-literature table (??), then the whole method is run on this paper’s own construction as a self-application demonstration (??). **Status.** All three propositions are carried as untested claims at tier DR, none above knowledge state K0; H3 is explicitly untested, and the self-application run itself surfaced real gate failures, not a clean pass (??, ??).

Abstract. People who live daily with a problem often know it before researchers do, but turning that knowledge into a checkable claim has historically required a university, a lab, or a research team. Generative AI removes that requirement technically while introducing a new one epistemically: it becomes cheap to produce text that reads as rigorous without it being clear which part is observation, which part is inference, and which part the AI supplied. This paper states, as three falsifiable propositions and not as demonstrated results, a candidate answer: a *claim card* that separately records what was seen, what the data separates, what AI filled in, what is assumed, and what independent check (if any) the claim has survived — bound to a formal Existence–Attribution–Disclosure norm and passed through a named independence ladder (I0 self-check through I5 outside human review) before public release — can carry the checkable property institutions have historically supplied, without requiring the institution itself (H1, design; H2, conceptual). A third, narrower proposition (H3, empirical) holds that cross-vendor AI checking measurably outperforms same-vendor re-checking at catching undisclosed AI-fill; it is stated but not tested here. Each proposition is placed against a literature conversation built from 48 independently verified citation cards, and the whole method is applied reflexively to this paper’s own construction, which surfaced real failures — a validator warning, a manifest-detected search-mode gap, and a documented citation-verification correction — reported as findings, not smoothed into a clean success story. Every claim here sits at tier DR and knowledge state K0; none has passed an independent (I5) human check.

Positioning. We are not competing with knowledge-authority and make no priority claim; we state what we propose, what we build on, and what would make us wrong.

1 PROBLEM BEFORE OBSERVATION

“Observation” is abstract; it is a verb, not a starting node. Founder’s own line, verbatim (Blackbox Log BBL-2026-09-04-083, concept DOI 10.5281/zenodo.22302518):

แต่คำว่าสังเกต มันเป็นนามธรรม คือเป็นกริยา ถ้าเราเริ่มจากปัญหาหาคะ มันมีกระบวนการแบบนี้นะ
เราค่อยเสนอให้โฟกัสปัญหาแทน

— “the word ‘observation’ is abstract, it is a verb; if we started from the *problem* instead, is there a process like this? We propose focusing on the problem first.”

A problematic state exists in the world *before* any knower acts on it — before it is looked at, named, or measured. The spine this paper’s claims are built on (adopted this pass into design/FOUNDATION_v0.5.md §2.1a, BBL-2026-09-04-084) therefore does not start with “Reality →” as its first node — the lens never grants direct access to reality, only to what was read out ($R_A = O_A(W; \Pi_A) \neq W$) — it starts one layer finer:

Problem state (unread) → first human readout (the Blackbox Note’s raw line, verbatim, untranslated) → human-language question → [lens-in] → hypothesis (lens-out, signed) → evidence → provisional knowledge (K0, ...)

This paper’s own problem state, before it was read out at all: a person who lives daily with a problem, has no university appointment, no certified lab, and no standing research team, and now has general-purpose AI assistance. The first human readout of that state, verbatim (Thai, source of truth), Blackbox Note line 1

(blackbox/BLACKBOX_NOTE_glosa-paper_2026-09-04.md#19)

...คนที่ไม่มีมหาวิทยาลัย ไม่มีเงิน และไม่มีทุนวิจัย จะผลิตความรู้จากพื้นที่ของตนเองอย่างเข้มงวดได้อย่างไร...

— “...how can a person with no university, no lab, and no research team produce rigorous knowledge from their own site of practice...”

The contaminated concept this question already flags: “rigour” ordinarily presupposes an institutional mechanism (a lab, a department, a review committee) sitting behind a claim, rather than naming a property a single claim can carry on its own, field by field.

2 STANDPOINT AND OWNERSHIP

This paper is written from the standpoint of a social-enterprise practitioner and citizen (the founder, Yaoharee Lahtee), not a substitute expert in computer science, philosophy of science, library/information science, or research-integrity studies. **Disciplines not claimed:** medicine, religious/fiqh authority, law, engineering, anthropology, political science, and formal computer science beyond what is mechanically executed here. This is disclaimer **D-STANDPOINT / D-NONEXPERT** in ??.

2.1 Responsibility per arrow

Every arrow on the spine above is named by who performed it. **Data** → **Inference** may be human, ai, or joint; the three project claim cards behind this paper (??) record it as joint — AI retrieved literature and drafted candidate hypotheses, the founder framed the problem and question. **Inference** → **Claim is always human**. AI is never an author. This is not a new rule so much as this pass naming, per arrow, a distinction the repo already held at the whole-artifact level (AGENTS.md gate rule 5).

2.2 Ownership: problem, question, and the selection of the hypothesis stay human

Founder’s line, verbatim, first stated (BBL-2026-09-04-086) and then self-corrected (BBL-2026-09-04-088, the correction shown here is the one adopted):

หรืออีกทางหนึ่งก็คือ ถ้าปัญหายังเป็นของเร่ คำถามยังเป็นของเร่ และ ‘การเลือก’ สมมุติฐานยังเป็นของเร่ แล้วถ้าไปสู่การแก้ปัญหาได้เร่ จะใช้อะไรใช้แนวไหนดี ก็ใช้ไปเถอะ เพราะมันคือเป้าหมายของความรู้อยู่ แต่ปัญหาก็ใคร่สักระยะ

— “if the problem is still ours, the question is still ours, and the *selection* of the hypothesis is still ours, and it leads to solving our problem, then use AI, use a cat, use a bear, go ahead — because that is the goal of knowledge: to solve a problem for someone.”

Tag	Meaning
TH_COQC	Machine-checked, axiom-free formal result.
FINITE_DIAGNOSTIC	Measured/computed on a finite, retained, reproducible input; not a limit or population claim.
FIT_CALIBRATED	A model calibrated against declared data; accuracy conditional on that fit.
DR	Declared bridge / human narrative synthesis; a design or interpretive judgment, not a proof or measurement.
DEFINITION	A stipulated definition used only where decision-bearing.
OPEN	Not established; a real open question, not smoothed over.

Table 1: Claim-tier legend (readout_genesis + readout_universe ladder). Every propositional claim in this paper (H1, H2, H3, ??) is DR; the demonstration (??) is FINITE_DIAGNOSTIC where a command was actually executed.

Three acts on the spine are the human’s: owning the **problem**, asking the **question**, and **selecting** the hypothesis. Every other node — retrieval, drafting, analysis, checking, rendering — may be performed by any tool. Concretely for this paper: AI (the AI assistant) drafted candidate hypotheses H1, H2, and H3 (hypotheses.md); the founder’s own act of selection, recorded in hypothesis_selection.yaml and confirmed by ruling BBL-2026-09-04-095 (Thai verbatim in entries.jsonl; gloss: “bring all three together”), chose all three — H1 and H3 are carried as untested propositions with stated falsifiers, and H2 is advanced in the paper’s own voice as its central conceptual claim (??, genre_decision.yaml).

3 THE LENS AND THE FIVE QUESTIONS

3.1 Claim-tier legend

Every claim in this paper carries exactly one tier tag; tiers are never collapsed or silently upgraded (??).

3.2 The readout-not-truth lens

This paper’s problem-to-hypothesis translation was performed under *Readout Universe — Yaoharee Lahtee* (lens; DOI 10.5281/zenodo.21529456, 10.5281/zenodo.21665100; repositories https://github.com/morrocwi/readout_universe, https://github.com/morrocwi/readout_genesis). Naming the lens credits its source and lets a reader replace it; it does not make the lens, or any hypothesis drafted under it, true (disclaimer **D-EXTERNAL-INPUT**). Everything read under this lens is a finite, retained *readout* of a source, never the source itself: a benchmark number, a citation’s exact passage, a claim card’s own field, this paper’s own prior draft — each was produced by some process, at some resolution, under some method, and is read as that readout, not as truth handed down.

3.3 Existence–Attribution–Disclosure and the five questions

The Readout Condition is $E \wedge A \wedge D$: a claim satisfying Existence (some provenance path for every load-bearing distinc-

#	Founder's question	Mechanically check-able?
Q1	<code>what_was_actually_seen</code> — what was actually seen	presence only
Q2	<code>what_the_data_separates</code> — what the data separates	presence only
Q3	AI <code>chunks</code> — what AI filled in	presence only
Q4	<code>what_is_assumed</code> — what is assumed	presence only
Q5	<code>independent_evidence_objection</code> — what independent evidence/objection	the one genuinely enforce-able check

Table 2: Founder's five questions $\Leftrightarrow$ E-A-D $\Leftrightarrow$ claim-card field group (`design/FOUNDATION_v0.5.md` §3.1). A card that answers Q1/Q2/Q4 honestly but misattributes Q3 — crediting the source with a distinction AI actually supplied — fails the Readout Condition even though every field looks filled; `silent_lift_check` is the mechanized test for exactly this.

tion), Attribution (identification of every non-source node), and Disclosure (every AI-supplied distinction named, not silently absorbed into the claim) is what a claim card is built to record, field by field, against the founder's own five questions (??).

Only Q5 — whether an independent check was ever run, and at what independence class — is structurally enforceable by the kernel; Q1–Q4 are presence-checkable, never correctness-checkable, because *mechanical validity* $\neq$ *semantic validity*. This is why the independence ladder (??, ??) is the licensed route to closing what the other four questions cannot mechanically close on their own.

4 GENRE AND EVIDENCE STANDARD

This paper is genre-routed as **CONCEPTUAL** by a human, adopted-by-human override of the router's own mechanical output (`genre_decision.yaml`). The router (run with `-search-log`) returned `systematic_review` because a frozen `search_log.yaml` exists under each of h1/h2/h3; the founder's ruling (BBL-2026-09-04-097, Thai verbatim in `entries.jsonl`; gloss: "make this document an academic article, a concept paper") does not adopt that output — a documented search log supporting three candidate hypotheses is not the same fact as the genre of the claim the paper itself advances, and the router has no way to see that distinction (`genre_decision.yaml` note). H2 (the conceptual, portable-property claim) is advanced in the paper's own voice; H1 (design) and H3 (empirical) are carried as propositions with stated falsifiers, neither run as a study this pass. Evidence standard: every citation is a status: `VERIFIED` card (??); every propositional claim stays DR until an independent test says otherwise.

5 THREE PROPOSITIONS

Candidates were drafted by AI from the lens translation above; the founder selected among them (??). All three are carried as propositions, not results.

H1 — Design: the claim-card mechanism (^[C1])

Statement. If every claim a standalone scholar produces is recorded as a claim card answering the five questions, tied to E-A-D, and passed through the independence ladder before any public release, then a person with no institutional affiliation can co-produce knowledge with AI whose rigour a reader can check for themselves, without an institutional credential serving as the certifying step.

Falsifier. A claim card that passes every field the kernel can mechanically check is nonetheless found by a genuine independent reviewer (I5) to have a false answer at Q2 or Q3 — showing the mechanical check does not track the epistemic property it claims to track.

What would count as evidence. A corpus of claim cards run through the independence ladder to at least I3, with a subset reaching genuine I5, compared against a matched corpus produced without the discipline, scored on whether an independent reader can recover the evidence/AI-fill boundary the maker recorded.

H2 — Conceptual: rigour as a portable claim-level property (^[C2])

Statement. Rigour is not a property institutional machinery confers on a claim from the outside; it is a property a single claim can carry in its own recorded structure — institutions are *one* historical mechanism that has produced this property, not the only possible one.

Falsifier. A claim card that answers all five questions honestly and completely, checked at I3, is shown to still require an institutional credential — not just an independent human reviewer — to be treated as rigorous by its actual intended readers, for at least one class of reader/decision this project cares about.

What would count as evidence. Reader studies on whether a target audience (journal editors, grant reviewers, a practitioner's own community) treats a well-formed claim card as sufficient without also asking who backs it.

H3 — Empirical: cross-vendor checking raises AI-fill detection (^[C3], untested)

Statement. When claim cards are checked by a different AI vendor from the one that produced them (I3) instead of the same vendor re-checking (I0/I1), the rate at which undisclosed AI-fill (a false Q3 field) or undisclosed assumptions (a false Q4 field) is caught rises measurably.

Falsifier. A controlled comparison (claim cards seeded with a known number of undisclosed AI-fill/assumption instances,

same-vendor vs. cross-vendor routes) shows no significant difference, or shows cross-vendor performs worse.

What would count as evidence. A seeded-error benchmark with a pre-registered scoring rubric; tiered FIT_CALIBRATED at best until run, DR until then. **This paper does not run that benchmark; H3 is stated, not tested.**

All three propositions were drafted under the same lens (??); none has an evidence relation above independence class I3, and none of the underlying claim cards has yet passed founder sign-off (`independent_check.status: NONE` on all three, ??).

6 CONVERSATION WITH THE LITERATURE

Each table below is a *conversation with the problem*, not a chronology or priority ranking (`design/S14_literature-review-system.md`): sources are placed by how they talk to the proposition, never by who came first. Every row resolves to a citation card with `status: VERIFIED` in the corresponding `litreview_manifest.yaml` (`records/lit/rigour-without-infrastructure/h{1,2,3}/`), meaning both `metadata_verified` (Cross-ref/OpenAlex/arXiv lookup) and `claim_match_verified` (a decorrelated cross-vendor AI route, `route:codex`, I3) hold. **Relation column** follows the repo’s comparison rule (same/different/cited, never a priority claim): *same* = the source’s own finding is compatible with the proposition’s direction; *different* = the source’s own finding counts against it; *cited* = the source bears on the topic without a same/different verdict (orthogonal or undetermined in the underlying dialogue table); rows marked *own lineage*, *not independent* are the founder’s own prior published pieces, collided against world literature per instruction BBL-2026-09-04-098 and never counted as independent support.

Across all three tables, 20 of 48 rows (42%) are the founder’s own prior published pieces, collided against world literature per instruction BBL-2026-09-04-098 rather than left silent; every own-lineage row is marked *own lineage*, *not independent* and none is counted toward any of the three propositions’ evidential support.

7 THE ARTIFACT APPLIED TO ITSELF

This section reports what the 2026-09-04 self-application run — glosa run on glosa’s own paper, per instruction BBL-2026-09-04-091 — actually recorded, as a readout of that run, not as a claim that the method passed cleanly.

Gate results actually executed

Running `./cli/glosa claim validate` directly against the three project claim cards (GLOSA-CC-20260904-0201/0202/0203) this session returns verdict: `PASS_WITH_LIMITS` for all three, with

zero schema errors, but with one identical warning on every card: “*D-LENS-UNCITED not checked – validate_claim_card was not given citation_cards (lens_translation.lens_ref is set); pass citation_cards=[...] to close this gap.*” The validator is honest about its own coverage limit here: a schema-level PASS is not, by itself, a claim that the lens-citation gate closed — it closed only once this session additionally cross-checked the lens DOIs against the citation-card apparatus described in ??.

Composite quotes caught by the cross-vendor route

The single strongest gate finding this run: an I3 (decorrelated cross-vendor) check on an earlier citation pass caught *composite quotes* — citations assembled from memory rather than a locator taken from the opened source. The founder’s ruling in direct response, verbatim (BBL-2026-09-04-100):

กำหนดไปเลยไหม ถ้าไปเอาข้อความต้องมีหน้าและบรรทัด
และสิ่งนี้มาครบห้ามเขียนจากคำนำแบบนั้นไปเลยดีกว่าจะได้ลดปัญหาตั้งแต่ต้นทางก่อนตรวจ

— “should we make it a hard rule: if a source is used it must carry a page and line locator plus a full link, never written from memory — that reduces the problem at the source, before the check.” Adopted immediately (BBL-2026-09-04-101, Thai verbatim in `entries.jsonl`; gloss: “update it into the system”), this became the rule under which every one of the 48 citation cards behind ?? was subsequently built: each carries `page_or_locator` and `fetched_from_url`, and each `exact_passage` is a source-first quotation, not a recalled paraphrase. This is a real, gate-caught failure that changed the method’s own rule (rule 17, `design/FOUNDATION_v0.5.md` §7.8), reported here because it happened, not because it flatters the method.

A manifest-detected search-mode gap

The genre router’s own mechanical output (??) returned `systematic_review` because `context.has_search_log` was true — but every one of h1/h2/h3’s own `litreview_manifest.yaml` records `search_mode` as `TARGETED_SEARCH` / `SCOPING_SEARCH`, never `SYSTEMATIC_REVIEW` — the manifest, read directly, is what let the founder catch that the router’s mechanical genre suggestion did not match the actual evidentiary standard the literature runs had met, and overrule it (??). This is the manifest doing exactly the job `design/FOUNDATION_v0.5.md` §7.8’s **FC-S8-1** hard block exists for: calling anything less than a full SR protocol a “systematic review” is blocked, here by the human reading the manifest against the router’s own claim.

Citation card	Source	Relation	Bearing
?	Mayes 2022, Citizen Science and Scientific Authority	different	Directly: even a perfectly-documented claim card may still not be treated as legitimate if the surrounding discourse continues to grant authority by credential.
?	Cwik 2026, Disclosure is not documentation	same	Separates disclosure-as-compliance from documentation-as-verifiable-record — the same separation H1 makes at Q3.
?	LLM hallucination survey, arXiv:2510.06265	cited	Shows AI cannot be trusted to flag its own errors — exactly why H1 requires an external check rather than AI self-report.
?	OCSDNET/HONF, Open Hardware and Citizen Science (Indonesia, Thailand, Nepal)	same	Grounds H1's premise that non-institutional knowledge production is a live regional practice, not a hypothetical.
?	Blackbox Log (Lahtee, Zenodo)	own lineage, not independent	Documents the append-only raw-record practice H1 assumes is followed; no claim about rigour by itself.
?	The Readout Condition (Lahtee)	own lineage, not independent	Supplies the E-A-D norm H1's claim card operationalizes; does not itself specify the card's fields.
?	Readout Genesis Standalone Synthesis (Lahtee)	own lineage, not independent	Supplies the independence-ladder vocabulary H1 tracks; does not itself define a claim-card schema.
?	The Standalone Scholar (Lahtee)	own lineage, not independent	Names the dual-track conversion (AI-assisted formation → independent human friction) H1's card is scoped inside, not a replacement for.
?	Written by AI. Still True. (Lahtee)	own lineage, not independent	Names "knower fetishism"; H1's card targets exactly the constitution-level property this paper argues authority derives from.
?	What a Zero Readout Certifies (Lahtee)	own lineage, not independent	Illustrates machine-checked discipline; H1's own mechanical checks (schema validation) are markedly weaker.
?	Kumar et al. 2025, Community perceptions of citizen science (South/SEA)	cited	Governance-inclusion framing, orthogonal to H1's rigour-checkability framing.
?	Jaeger et al. 2023, An epistemology for democratic citizen science	same	Locates robustness in diversity of standpoint/process — could argue H1 over-individualizes rigour to one standalone scholar.
?	Soderberg, Errington & Nosek 2020, Credibility of preprints	same	Researchers rate content/process cues over prestige when prestige is absent — supports the premise, not H1's specific design.
?	Scholarly Kitchen 2026, The Guild and the Gap	cited	A real rejection episode on prior-publication grounds; whether a claim card would change the outcome is untested.
?	TCIJ (TH) 2020, freelance-academic labour piece	cited	Adjacent labour/credit question, thinner on H1's actual rigour-verification topic than its search match suggested.

Table 3: H1 (design proposition) against 15 verified citation cards, 5 of them own-lineage. Diversity audit (`h1/litreview_manifest.yaml`): 1 Thai / 5 English external sources (own-lineage excluded), stance counts *agrees*:4 *orthogonal*:2 *undetermined*:9 by the manifest's own collapsing convention (?? notes a discrepancy between this convention and the citation cards' own explicit notes). `concentration_flags`: []; `zero_disagree_flag`: false.

A stance-counting discrepancy this session found and did not paper over

Building ???? directly from each citation card's own notes field ("Dialogue-table stance: NO (disagrees with / challenges H1)" on `cite-cstp-boundary-work-001`) surfaced a discrepancy with the corresponding manifest's own `diversity_audit.counts.by_stance`: the manifests for H1 and H3 both record `disagrees: 0`, while this session confirmed directly (`grep` against the raw dialogue-table rows, `FINITE_DIAGNOSTIC`) that at least one row per hypothesis (Mayes 2022 against H1; Xiang et al. 2026 against H3) is an explicit, single-value challenge row that a card's own notes field independently labels a disagreement. The manifest's aggregate counting convention appears to collapse a dual-value "NO/NO" dialogue-table cell into undetermined rather than disagrees, undercounting real challenges in its own summary statistic by at least one row per hypothesis where this occurs. This paper's own tables (???) use the citation card's explicit notes field, not the manifest's aggregate count, and this discrepancy is disclosed rather than silently reconciled — a small instance of exactly the "mechanical check does not track the epistemic property it claims to track" failure mode H1's own falsifier names, found here in the tool that mechanically summarizes H1's own supporting literature.

Path leaks caught by CI, not by this run

The repository's public-facing leak gate (`scripts/check_leak.sh`, wired into `.github/workflows/ci.yml`'s leak-scan job) is a repo-level, not a paper-level, control: it did not need to run specifically against this paper's files this session (none of this paper's content matched a denylisted pattern), but it is the mechanical backstop this paper's own public-facing disclosures rely on, and its presence in CI — rather than only as a manual pre-publish step — is itself part of what this paper's demonstration reports: a leak gate that runs on every push, not only when someone remembers to ask for it.

8 LIMITATIONS AND DISCLAIMERS

D-STANDPOINT

See ??. This paper is written from a social-enterprise-practitioner/citizen standpoint, not disciplinary authority; a reader who skips to this section should read every claim above as offered from that position and access level, never from institutional credential.

D-NONEXPERT

The author is not a substitute for a credentialed expert in citizen-science studies, philosophy of science, library/information science, or empirical AI-evaluation methodology — the disciplines this paper draws on without claiming standing in. Medicine, religious/fiqh author-

Citation card	Source	Relation	Bearing
?	Wikipedia, Open-notebook science	cited	Treats openness as a property of record-keeping practice, close to but coarser than H2's own five-field structure.
?	Rasmussen 2019, Confronting Research Misconduct in Citizen Science	cited	Calls for institution-like misconduct-detection machinery — cuts against H2's claim that the property needs no such mechanism.
?	Freitag, Meyer & Whiteman 2016, credibility-building strategies	same	Credibility built bottom-up through concrete recorded practices, not top-down certification.
?	Scholarly Kitchen 2026, The Guild and the Gap (+ comments)	different	Close to H2's own falsifier: a named commenter states affiliation/publication history earns <i>more</i> scrutiny than any single record could substitute for.
?	Dutta et al. 2021, Decolonizing Open Science	cited	Structural/platform-governance framing, adjacent to but different from H2's single-claim scope.
?	Okafor et al. 2022, Institutionalizing Open Science in Africa	cited	Aims for <i>more</i> institutional machinery as the path forward — the opposite emphasis from H2, at a different level.
?	Kumar et al. 2025, community perceptions of citizen science	cited	Access to institutional audiences, not record quality, is the bottleneck this source finds — adjacent friction H2 does not resolve.
?	Srirrot & Sohomboon 2025 (TH), qualitative Trustworthiness framework	same	Record-level techniques (audit trail, triangulation) let a reader judge rigour without institutional standing — the closest structural match to H2 found.
?	THE STANDARD (TH) 2025, Thai Open Science policy piece	cited	Institution-led opening of an existing record — the institution still decides what counts as legitimate, an access question not a rigour-provenance one.
?	Blackbox Log (Laatee)	own lineage, not independent	No claim about whether rigour is portable at claim level.
?	The Civilization of Knowledge (Laatee)	own lineage, not independent	Locates the stabilizing step historically in institutions — would ask whether a claim card is a new non-institutional stabilizer or a mistaken substitute.
?	The Readout Condition (Laatee)	own lineage, not independent	A claim satisfying E-A-D does not need an institution to vouch for it; does not itself argue institutions add nothing.
?	Readout Genesis Standalone Synthesis (Laatee)	own lineage, not independent	Institutions are one belief-scale among several, not a precondition for knowledge status.
?	The Standalone Scholar (Laatee)	own lineage, not independent	Independent human friction is one specific historical route to stress-tested knowledge, not necessarily the only one — would ask if H2 still needs some friction step.
?	Written by AI. Still True. (Laatee)	own lineage, not independent	Institutional authority is derivative, not a source, of epistemic value — the same possession/constitution distinction H2 draws.
?	What a Zero Readout Certifies (Laatee)	own lineage, not independent	Illustrates zero-institutional-apparatus rigour in a formal setting — weak indirect support only.

Table 4: H2 (conceptual proposition, this paper's own advanced claim) against 16 verified citation cards, 6 own-lineage. Diversity audit (h2/litreview_manifest.yaml): 3 Thai / 13 English, stance *agrees*:6 *disagrees*:2 *orthogonal*:7 *undetermined*:1. `concentration_flags`: []; `zero_disagree_flag`: `false` — this is the one proposition of the three whose manifest itself records a nonzero disagree count.

ity, law, engineering, anthropology, political science, and formal computer science beyond what is mechanically executed here are explicitly not claimed.

D-SCOPE

Not applicable in the empirical-population sense (this is a CONCEPTUAL genre paper, ??). Where a source's own sample size or population matters to a comparison row (?????), it is stated inline in that row rather than in one blanket population statement.

D-NONCLAIM

This paper does *not* claim: that H1's mechanism has been run as a study; that H2's conceptual-portability claim has been tested against real readers; that H3's seeded-error benchmark has been built or executed; that the 48 citation cards behind ?????? constitute a systematic or exhaustive review (`search_mode`: `TARGETED_SEARCH` / `SCOPING_SEARCH`, never `SYSTEMATIC_REVIEW`); that any finding here is a medical, legal, or religious determination; or that cross-vendor AI checking (I3) is equivalent to, or a substitute for, an I5 human review.

D-AIFILL

See ??. AI (the AI assistant, "the AI assistant" seat) drafted candidate hypotheses H1/H2/H3, the literature-review runs, the claim cards' `current_evidence`, `retrieved_tool_evidence`, `retained_record_route`, `model_calibration_assumption`,

`prompt_system_constraint`, and `decision_policy` fields, and this paper's own prose and $\text{\LaTeX}$ build. None of this is treated as evidence on its own say-so (??).

D-TIER

The single highest tier claimed anywhere in this paper is DR (declared bridge / human narrative synthesis over a specified mechanism), with one `FINITE_DIAGNOSTIC` exception: the mechanically executed checks reported in ?? (schema validation, disclaimer emission, manifest gate reads) and this paper's own $\text{\LaTeX}$ compile. It cannot go higher because none of H1/H2/H3 has been run as a study, and no claim card behind this paper has passed a class-5 (I5) independent human check.

D-INDEPENDENCE

Every citation card behind ?????? was checked at independence class I3 (decorrelated cross-vendor AI route, `claim_match_verified_by`: `route:codex`) for `claim_match`, and mechanically (I3-adjacent, `Crossref/OpenAlex` lookup) for `metadata_verified`. The three project claim cards themselves (??) sit at `independent_check.status`: `NONE` — drafted and self-checked by the authoring AI session, never yet reviewed by a route independent of that session (this is I0 for the cards' own content, not to be conflated with the I3 citation-level checks that feed them). No external or institutional validation is sought or treated as available as a lever that would make any claim here more true; an outside

Citation card	Source	Relation	Bearing
?	Wataoka, Takahashi & Ri 2024/25, Self-Preference Bias in LLM-as-a-Judge	cited	Bias tracks perplexity/familiarity, not narrowly vendor identity — the effective variable may not always track vendor boundaries.
?	Xiang et al. 2026, Cross-Model LLM Code Review, arXiv:2607.21656	different	The one source that actually runs H3's own comparison: the effect is asymmetric and direction-dependent (one direction helps a lot, the reverse hurts), and same-vendor re-review also improved one condition.
?	Tan et al. 2025, Too Consistent to Detect (EMNLP)	same	Same-model repeated sampling cannot catch “self-consistent” errors — structurally the concern H3 raises about same-vendor re-check.
?	Du et al. 2023, Multiagent Debate	cited	Multi-instance debate improves factuality without varying vendor identity — the active ingredient could be plurality, not cross-vendor difference.
?	Huang et al. 2023/24, LLMs Cannot Self-Correct Reasoning Yet	cited	Tests zero-external-signal correction, a weaker condition than a same-vendor re-check with a fresh adversarial prompt — an imperfect analogy, flagged as such.
?	John, Singh & Panachakel 2026, Can Legal AI Know When It Is Wrong?	cited	Three vendors' error rates differ nearly 5x on identical inputs, consistent with H3's premise, but the paper's own lever for improved catching is human experience, not cross-vendor AI review.
?	Sultana et al. 2026, Bangladesh AI Readiness	cited	Names a capacity defeater of H3's practical reach (infrastructure to run any independent review at all), not of H3's empirical claim.
?	Fong 2026, AI Governance Policies in Southeast Asia	cited	Institutional-capacity constraint on deploying cross-vendor review as policy, orthogonal to H3's mechanism claim.
?	Cofact Thailand 2025, AI-image-detection reliability	cited	Adjacent domain (detecting whether content is AI-made, not whether AI-authored claims are honest) — genuinely different problem.
?	MindStudio blog 2026, Cross-Vendor AI Agent Review	same	States H3's own mechanism in practitioner language, but with no measurement, sample, or method of its own — flagged as concentration risk, not equivalent-weight evidence.
?	Blackbox Log (Lahtee)	own lineage, not independent	No claim about detection rates.
?	Knowledge as Stabilized Translation (Lahtee)	own lineage, not independent	Stability under variation is the marker of knowing well — conceptually the same move H3 makes; does not specify vendor identity as the variation dimension.
?	The Readout Condition (Lahtee)	own lineage, not independent	Disclosure is structural, not optional; does not specify the auditor must be a different AI vendor.
?	Readout Genesis Standalone Synthesis (Lahtee)	own lineage, not independent	Does not separate same-vendor from cross-vendor checking as a variable.
?	The Standalone Scholar (Lahtee)	own lineage, not independent	Independent validation in this paper's stronger sense requires human friction — cross-vendor AI (I3) is a weaker approximation.
?	Written by AI. Still True. (Lahtee)	own lineage, not independent	Reinforces that a checker's identity/route matters to what its check can license — no detection-rate comparison.
?	What a Zero Readout Certifies (Lahtee)	own lineage, not independent	Machine-checked verification, where available, is a strictly stronger independence route than cross-vendor AI review.

Table 5: H3 (empirical proposition, explicitly untested) against 17 verified citation cards, 7 own-lineage. Diversity audit (h3/litreview_manifest.yaml): 1 Thai / 6 English, stance *agrees*:3 *orthogonal*:4 *undetermined*:10 by the manifest's own convention. `concentration_flags: []`; `zero_disagree_flag: false`.

reader's input, if it comes, is optional-level evidence at whatever independence class it actually reaches, never a certification gate.

D-DVP-NOT-K2

A decorrelated-vendor AI review pass (DVP) is at most an *I3* route. **DVP** $\neq$ **K2**: no combination of AI-only checks, however many vendors, raises this paper's claims to K2. This paper's K-state is K0 (??); no claim in it is asserted above K0/DR.

D-LEGAL-NEQ-EPISTEMIC

None identified — no license, permit, or regulatory approval is referenced by this paper's claims. The general rule still holds: were one ever cited, it would make an action lawful, never raise epistemic warrant or certify a claim as true.

D-NOT-DIAGNOSTIC

Not applicable — this paper makes no health, clinical, or treatment recommendation of any kind.

D-TRANSLATION

Not applicable: per founder ruling (BBL-2026-09-04-099, ??), this concept paper is published in English only, with no Thai edition. Every founder line quoted in ?? nonetheless stays Thai, verbatim, untranslated in place, with an English gloss alongside it — the English-only rule governs the paper's own prose, not the Blackbox Note's raw lines.

Within the underlying claim cards, the Thai statement fields themselves carry `translation_status: reviewed`.

D-DISSENT-PRESERVED

None recorded to date for H1/H2/H3 specifically. The literature-review dialogue tables (????) already carry the closest real dissent this project has found against its own hypotheses (the Mayes 2022 boundary-work finding against H1 (?); the Scholarly Kitchen credentialism case against H2 (?)); these are stated as open challenges in ??, not resolved or smoothed over.

D-COMPARISON

This paper does not claim priority or precedence. ?? states every source relation as *agrees*, *disagrees*, or *orthogonal/undetermined* — never as a priority contest, never as “no other work does X.”

D-CITATION-UNVERIFIED

None identified for the 48 citation cards cited in ?????? — every one carries `status: VERIFIED` (`metadata_verified: true`, `claim_match_verified: true`) in its litreview manifest as of 2026-09-04; none is cited from memory (??, BBL-2026-09-04-100/101). Where a source's own full text was not fetched (e.g. `fetch_status: PAYWALLED_ABSTRACT_ONLY`), the citation card's scope field names the resulting evidentiary weakness

(PARAPHRASE rather than DIRECT_QUOTATION) inline in the relevant table row rather than silently upgrading it.

D-BLACKBOX-NOTE

This paper carries the mandatory Blackbox Note appendix (??). Note id: BB-2026-09-04-01 (projects/GLS-2026-001_rigour-without-infrastructure/blackbox_note.yaml), curated public source: blackbox/BLACKBOX_NOTE_glosa-paper_2026-09-04.md, public log entries BBL-2026-09-04-083, -086, -088, -089, -091, -095, -097, -098, -099, -100, -101 (blackbox/log/entries.jsonl, concept DOI 10.5281/zenodo.22302518).

D-K-STATE

Overall K-state: K0 (internal working note). Every load-bearing claim in this paper — H1, H2, H3, and every finding in ?? — shares this same K0 state; there is no per-claim exception. This matches the three underlying project claim cards' own k_state: K0 (??).

D-AUTHORSHIP

See ??. The problem, the research question, and the selection among H1/H2/H3 are the founder's own act (??); AI drafted every candidate, citation search, and this paper's prose, disclosed per claim (produced_by, ??), and never substitutes for the author's judgment or responsibility (**AIContribution** ≠ **EpistemicResponsibility**).

D-REVISION-LIVE

The three underlying claim cards (GLOSA-CC-20260904-0201/0202/0203) are at revision 2 of an actively edited project; status: Draft and independent_check.status: NONE on all three reflect that the founder has not yet signed off on any specific card instance, even though the founder has already ruled on genre (BBL-097) and hypothesis selection (BBL-095) at the project level.

D-NO-EPISTEMIC-VETO

No agency — including any AI vendor, reviewer, or institution named in this paper — holds an epistemic veto over H1, H2, or H3 merely by title, credential, or infrastructure ownership; any agency may propose, use, test, challenge, fork, translate, revise, restrict, or withdraw this work without needing permission, provided lineage is preserved (per the repo's own knowledge-validation stance, CLAUDE.md/AGENTS.md).

D-CANDIDATE-STATUS

H1, H2, and H3 are candidate hypotheses drafted by AI; the act of selecting among them (all three, per BBL-2026-09-04-095) was the founder's alone (??). No claim card here asserts candidate status was itself a form of testing.

D-EXTERNAL-INPUT

The lens this paper's readout-translation and hypothesis-drafting steps were performed under is named, cited, and linked throughout (??); naming it credits its source and lets a reader replace it — it does not make the lens, or any hypothesis drafted under it, true.

8.1 Knowledge state

This paper's overall knowledge state is **K0** (internal working note): every load-bearing claim in it, including ??'s findings, shares this K0 state. It is not K1 (public-provisional-with-a-named-independence class per claim and no class-5 human check yet) because the three underlying claim cards' own status field is still Draft and independent_check.status: NONE — the founder has ruled on genre and selection at the project level but has not yet signed off on any specific card instance (??).

9 AI-ASSISTANCE DISCLOSURE

This paper is a human-AI co-production, not AI-only automation and not human-only authorship. A generative-AI assistant (vendor recorded in the local cooking log, not named here per gate rule 9) assisted with literature retrieval and card drafting, the three claim cards' field content, the conversation-with-the-literature tables, and this L^AT_EX build. **AI output is not treated as evidence.** Every claim above is accepted only when it matches a retained source record (a citation card's exact_passage, the Blackbox Log's own entries, a mechanically executed command's own output) or a machine-checkable result reported in ??. The human holds the standpoint, the falsifier judgment, verification of load-bearing citations at the founder's own discretion, and the final public commitment (**AIContribution** ≠ **EpistemicResponsibility**).

10 WHAT WOULD CHANGE OUR MIND

Per hypothesis, stated plainly rather than left implicit:

- **H1** would be abandoned if a genuine I5 reviewer, working from a schema-passing claim card, found a false answer at Q2 or Q3 — showing the mechanical check gives false reassurance rather than tracking the property it names. It would be strengthened, not proven, by a corpus comparison at I3+ (with a subset at I5) against a matched non-card corpus.
- **H2** would be abandoned if a reader study found that credentialed audiences (editors, grant reviewers, a practitioner's own community) still required an institutional signal even after being shown a complete, well-formed claim card — i.e. the credential doing load-bearing work the card's fields could not substitute for. The Scholarly Kitchen credentialism case (cite-rigour-without-infrastructure-004) is already the closest real instance of exactly this challenge and is not explained away.
- **H3** is untested; the one source that actually ran its comparison (cite-crossmodel-codereview-002) found a direction-dependent, asymmetric effect rather than a uniform cross-vendor-beats-same-vendor pattern. A seeded-error benchmark that reproduces that asymmetry at the claim-card Q3/Q4 grain would be evidence against H3 as a general, direction-agnostic claim; a benchmark that finds a

uniform, direction-independent rise would be evidence for it.

11 CONCLUSION

This paper states three falsifiable propositions about whether a claim-card discipline can substitute for institutional infrastructure in making a standalone scholar’s claim reader-checkable, places each against verified literature rather than memory or silence, and then runs the same discipline on its own construction — finding, honestly, a validator warning, a caught set of composite quotes that changed the method’s own citation rule mid-project, a router/manifest mismatch the founder had to catch by hand, and a small stance-counting discrepancy in the project’s own summary statistics. None of this is smoothed into a clean success story: every claim here stays DR, every underlying card stays K0 and unsigned, and H3 stays explicitly untested. What is offered is not a demonstrated result but a specified mechanism, an honest account of what happened when it was pointed at itself, and a named falsifier for each of the three ways it could be shown wrong.

A CLAIM MATRIX

Per Chair Ruling v1 §B2, this is a print-space compression of the three full claim cards (projects/GLS-2026-001_rigour-without-infrastructure/claims/GL0SA-CC-20260904-0201/0202/0203.yaml); field-presence here is mechanically checkable, correctness is what the independent check (Q5) exists for.

A.1 Five-questions detail

B BLACKBOX NOTE: HOW THIS WORK WAS MADE

(*Thai project term: “Bantuek Klong Dam”*.) Curated verbatim lines — typos kept, never polished or translated in place — followed by the cooking log. Full record archived separately (projects/GLS-2026-001_rigour-without-infrastructure/blackbox_note.yaml, note id BB-2026-09-04-01; public curated source blackbox/BLACKBOX_NOTE_glosa-paper_2026-09-04.md; public log entries cited by id below, blackbox/log/entries.jsonl, concept DOI 10.5281/zenodo.22302518).

C HUMAN MASTERY GATE

Before submission the author confirms they can defend, without AI assistance: (1) the problem (??) (2) the major literature families (??) (3) the closest constructs (own-lineage rows, ??) (4) the precise inadequacy (the contaminated concept named in ??) (5) the proposed mechanism (H1, ??) (6) the boundary conditions ($E \wedge A \wedge D$, ??) (7) the strongest counterargument (the Scholarly Kitchen credentialism case, ??) (8) the strongest counterexample (the direction-dependent cross-model result

against H3, ??) (9) the falsifying record for each proposition (??) (10) the discriminating empirical test (H3’s not-yet-run seeded-error benchmark). **Gate status: PASS WITH NAMED GAPS** — the founder has not yet performed the class-5 (I5) sign-off on any of the three underlying claim cards (??); this gap is named, not silently dropped.

D PROVENANCE AND LINEAGE

Status vocabulary: PRESERVE_EXACT, PRESERVE_FUNCTION, EXPAND, SUPERSEDE. This paper PRESERVE_FUNCTIONS the claim-card schema and independence ladder from design/FOUNDATION_v0.5.md and EXPANDs it with a three-proposition literature conversation and a self-application demonstration specific to this project (GLS-2026-001). Direct ancestor: the Standalone Scholar dual-track architecture (?). Core epistemic engine this paper’s disclosure apparatus is built on: the Readout Condition (?).

ID	Informal statement	Evidence / support	Tier	Produced by	Responsible
C1	H1: claim-card + E-A-D + independence ladder lets a standalone scholar's claim be reader-checkable without institutional credential.	15 verified citation cards (??); GL0SA-CC-20260904-0201	DR	joint (data→inference); human (inference→claim)	human
C2	H2: rigour is a portable claim-level property, not conferred solely by institutional machinery.	16 verified citation cards (??); GL0SA-CC-20260904-0202	DR	joint (data→inference); human (inference→claim)	human
C3	H3: cross-vendor (I3) checking measurably outperforms same-vendor (I0/I1) checking at catching undisclosed AI-fill. Untested.	17 verified citation cards (??), all analogical, none a direct test of Q3/Q4 detection; GL0SA-CC-20260904-0203	DR	joint (data→inference); human (inference→claim)	human

Table 6: Claim Matrix. All three cards: `k_state`: K0, `status`: Draft, `independent_check.status`: NONE, `independence_class`: I3 (the citation-verification route, not the card's own sign-off).

ID	seen	ai_filled	assumed	tested (indep.)
C1	Blackbox Note L1; founder's abstract framing	candidate hypothesis text, evidence retrieval, card drafting (all disclosed, <code>disclaimers_emitted</code> : D-AIFILL)	no source tests the full combined H1 mechanism as one system	I3 (citation checks); card itself unsigned
C2	same L1; H2's own narrower conceptual framing	same AI-fill fields as C1	the two SUPPORTS sources are same-lineage; cannot alone move H2 past DR	I3; card itself unsigned
C3	same L1; H3's cross-vendor detection-rate framing	same AI-fill fields as C1	no seeded benchmark has been run; the closest analogous study found a direction-dependent effect	I3 (analogical citations only); card itself unsigned

Table 7: Five-questions detail keyed to ??. Full cards: `projects/GLS-2026-001_rigour-without-infrastructure/claims/GL0SA-CC-20260904-0` (and -0202, -0203).

ID	Kind	Thai verbatim	Distinction introduced	Became
BBL-083	proposal	see ?? image above	Separates “observation” (abstract verb) from “problem state” (concrete pre-existing thing)	FOUNDATION §2.1a (??)
BBL-086/088	finding / correction	see ?? image above	Which spine nodes must stay human: problem, question, and the <i>selection</i> of the hypothesis (088 corrects 086)	FOUNDATION §2.1c
BBL-089	ruling	<code>entries.jsonl</code> (not reproduced here; column-width)	Ratifies the finer-grained spine diagram (AI-drafted wording, human-ratified agreement)	FOUNDATION §2.1a diagram
BBL-091	instruction	<code>entries.jsonl</code> (not reproduced here)	Install and use the skill to build this article in full	task B1/B2 launch
BBL-095	ruling	<code>entries.jsonl</code> (not reproduced here)	Selects all three hypotheses (H1, H2, H3) together	<code>hypothesis_selection.yaml</code>
BBL-097	ruling	<code>entries.jsonl</code> (not reproduced here)	Sets genre to CONCEPTUAL, overriding the router's <code>systematic_review</code> output	<code>genre_decision.yaml</code> (??)
BBL-098	instruction	<code>entries.jsonl</code> (not reproduced here)	Requires colliding own prior papers against world literature, not silence	own-lineage rows, ??
BBL-099	ruling	<code>entries.jsonl</code> (not reproduced here)	English only, no Thai edition; arXiv two-column theme, polished	this document's language and theme
BBL-100	ruling	see ?? image above	Mandates locator+link, source-first citation, after I3 caught composite quotes	rule 17, §7.8; ??
BBL-101	instruction	<code>entries.jsonl</code> (not reproduced here)	Adopt the citation rule into the system immediately	rule 17 adoption

Table 8: Blackbox Note — curated verbatim lines relevant to this paper's own construction (full Thai retained; English gloss given inline in the surrounding text per section, not translated here in place).

Step	By	Input lines	Output	What changed
lens_in	ai-assistant	L1	<code>lens_translation.md</code>	Drafted <code>question_readout</code> , <code>local_contrast_space_X</code> , <code>access_relation_R</code> , <code>claim_function_Phi_z0</code> from L1 under the Readout Universe lens; not yet independently reviewed.
lens_out	ai-assistant	L1–L13	<code>hypotheses.md</code>	Drafted H1/H2/H3 as candidate world-language hypotheses; selection reserved to and performed by the founder (BBL-095).
lit_review	ai-assistant	H1/H2/H3	48 citation cards, 3 <code>litreview_manifest.yaml</code>	Ran targeted/scoping searches per hypothesis (not systematic), froze manifests, computed diversity/accuracy audits.
claim_card	ai-assistant	hypotheses + literature	GL0SA-CC-20260904-0201/0202/0203	Filled full-shape cards; validated PASS_WITH_LIMITS; not yet founder-signed.
paper_build	ai-assistant	all of the above	this document	Drafted the concept paper’s prose, tables, and $\LaTeX$ build; disclosed throughout, never authored.

Table 9: Cooking log (Thai project term: “karn pung”) — append-only.